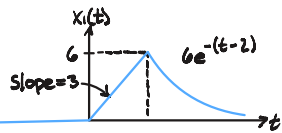


$$x(t) = e^{j\omega t}$$

$$y(t) = \hat{H}(\omega) e^{j\omega t}$$

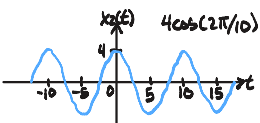
Energy signals:

$P_{av} = 0$, E is finite



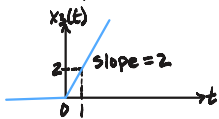
Power signals:

P_{av} is infinite, E is infinite



Non-physical signals:

Both P_{av} and E are infinite



$$\frac{dx}{dt} = j\omega x e^{j\omega t}$$

$$\frac{dy}{dt} = j\omega \hat{H}(\omega) e^{j\omega t}$$

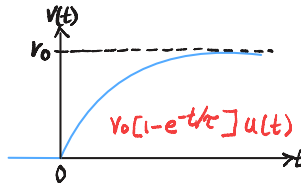
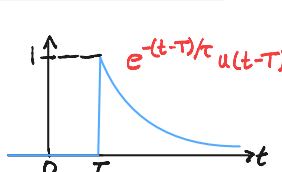
$$\frac{d^2y}{dt^2} = (j\omega)^2 \hat{H}(\omega) e^{j\omega t}$$

Impulse response

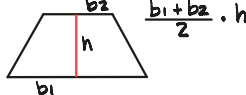
$$y(t) = \frac{1}{RC} e^{-t/RC} u(t) = h(t)$$

$$\cos \omega t = \frac{1}{2} [e^{j\omega t} + e^{-j\omega t}]$$

$$\sin \omega t = \frac{1}{2j} [e^{j\omega t} - e^{-j\omega t}]$$



trapezoid:



Integration by parts

$$\int u dv = uv - \int v du$$

Discriminant Notes

$$D > b^2 - 4ac$$

$D > 0$ 2 real roots

$D < 0$ 2 complex roots

$$p_1 = \frac{-a_1 + \sqrt{a_1^2 - 4a_2}}{2}$$

$$p_2 = \frac{-a_1 - \sqrt{a_1^2 - 4a_2}}{2}$$

$$p_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = \omega_0 \left[-\xi + \sqrt{\xi^2 - 1} \right]$$

$$\Rightarrow p_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = \omega_0 \left[-\xi - \sqrt{\xi^2 - 1} \right]$$

$$\alpha = \frac{a_1}{2} \quad \omega_0 = \sqrt{a_2} \quad \xi = \frac{\alpha}{\omega_0}$$

(Np/s) (rad/s) no unit

Transformation

Expression

Consequence

Time shift

$$y(t) = x(t-T)$$

Wave form is shifted by T

Time scaling

$$y(t) = x(at)$$

Wave is compressed if $|a| > 1$

Wave is expanded if $|a| < 1$

Time reversal

$$y(t) = x(-t)$$

Wave form is mirrored across

Generalized

$$y(t) = x(at-b)$$

the vertical axis

scale by a , then time by $1/a$, time reversal $a < 0$

even/odd synthesis $x(t) = x_e(t) + x_o(t)$

$$x_e(t) = \frac{1}{2} [x(t) + x(-t)] \quad x_o(t) = \frac{1}{2} [x(t) - x(-t)]$$

Linear differential equations

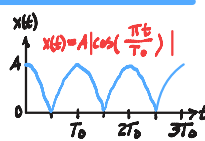
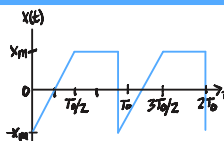
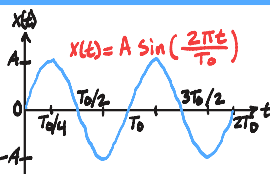
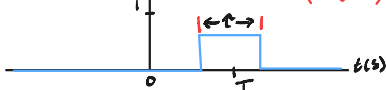
$$\frac{dy}{dx} + P(x)y = Q(x)$$

Integrating factor: $e^{\int P(x) dx}$



$$h(t) = h_1(t) * h_2(t) + h_3(t) * h_4(t)$$

rectangular pulses $\text{rect}\left(\frac{t-T}{\tau}\right)$



$$x(t) = A \cos(\omega_0 t + \theta), -\infty < t < \infty$$

A = Amplitude; $x_{max} = A$ and $x_{min} = -A$

ω_0 = Angular frequency in rad/s

θ = phase shift, $f_0 = \omega_0 / 2\pi$ (Hertz)

$T_0 = 1/f_0 = \frac{2\pi}{\omega_0}$ Period of $x(t)$ in seconds

even x even = even

even x odd = odd

odd x odd = even

$$\pi/6 = 30^\circ = (\sqrt{3}/2, 1/2)$$

$$\pi/4 = 45^\circ = (\sqrt{2}/2, \sqrt{2}/2)$$

$$\pi/3 = 60^\circ = (1/2, \sqrt{3}/2)$$

Table 2.1: convolution properties

Convolution Integral

$$y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau$$

Property

Description

1. Commutative

$$x(t) * h(t) = h(t) * x(t)$$

2. Associative

$$[g(t) * h(t)] * x(t) = g(t) * [h(t) * x(t)]$$

3. Distributive

$$x(t) * [h_1(t) + \dots + h_N(t)] = x(t) * h_1(t) + \dots + x(t) * h_N(t)$$

4. Causal * Causal = Causal

$$y(t) = u(t) \int_0^t h(\tau) x(t-\tau) d\tau$$

5. Time-shift

$$h(t-T_1) * x(t-T_2) = y(t-T_1-T_2)$$

6. Convolution with Impose

$$x(t) * \delta(t-T) = x(t-T)$$

7. Width

Width of $y(t)$ = Width of $x(t)$ + Width of $h(t)$

8. Area

Area of $y(t)$ = Area of $x(t)$ + Area of $h(t)$

9. Convolution with $u(t)$

$$y(t) = x(t) * u(t) = \int_{-\infty}^t x(\tau) d\tau \text{ (Ideal integrator)}$$

10a. Differentiation

$$\left(\frac{d^m x}{dt^m}\right) * \left(\frac{d^n h}{dt^n}\right) = \frac{d^{m+n} y}{dt^{m+n}}$$

10b. Integration

$$\int_{-\infty}^t y(\tau) d\tau = x(t) * \left[\int_{-\infty}^t h(\tau) d\tau\right] = \left[\int_{-\infty}^t x(\tau) d\tau\right] * h(t)$$

Commonly encountered convolutions

1) $u(t) * u(t) = t u(t)$

2) $e^{at} u(t) * u(t) = \left(\frac{e^{at}-1}{a}\right) u(t)$

3) $e^{at} u(t) * e^{bt} u(t) = \left[\frac{e^{at}-e^{bt}}{a-b}\right] u(t) \quad a \neq b$

4) $e^{at} u(t) * e^{at} u(t) = t e^{at} u(t)$

5) $t e^{at} u(t) * e^{bt} u(t) = \frac{e^{bt}-e^{at} + (a-b)t e^{at}}{(a-b)^2} u(t)$
 $a \neq b$

6) $t e^{at} u(t) * e^{at} u(t) = \frac{1}{2} t^2 e^{at} u(t)$

7) $\delta(t-T_1) * \delta(t-T_2) = \delta(t-T_1-T_2)$

Trigonometric Identities

Even and Odd Formulas

$$\sin(-\theta) = -\sin \theta$$

$$\cos(-\theta) = \cos \theta$$

$$\tan(-\theta) = -\tan \theta$$

Product to Sum Formula

$$\cos x \cos y = \frac{1}{2} [\cos(x-y) + \cos(x+y)]$$

$$\sin x \cos y = \frac{1}{2} [\sin(x+y) + \sin(x-y)]$$

$$\sin x \sin y = \frac{1}{2} [\cos(x-y) - \cos(x+y)]$$

$$\cos x \sin y = \frac{1}{2} [\sin(x+y) - \sin(x-y)]$$

Half Angle Formulas

$$\sin\left(\frac{a}{2}\right) = \pm \sqrt{\frac{1-\cos a}{2}}$$

$$\cos\left(\frac{a}{2}\right) = \pm \sqrt{\frac{1+\cos a}{2}}$$

$$\tan\left(\frac{a}{2}\right) = \pm \sqrt{\frac{1-\cos a}{1+\cos a}}$$

$$= \frac{\sin a}{1+\cos a} = \frac{1-\cos a}{\sin a}$$

Sum to Product

$$\sin x + \sin y = 2 \sin \frac{(x+y)}{2} \cos \frac{(x-y)}{2}$$

$$\sin x - \sin y = 2 \sin \frac{(x-y)}{2} \cos \frac{(x+y)}{2}$$

$$\cos x - \cos y = -2 \sin \frac{(x+y)}{2} \sin \frac{(x-y)}{2}$$

$$\cos x + \cos y = 2 \cos \frac{(x+y)}{2} \cos \frac{(x-y)}{2}$$

Double Angle Formulas

$$\sin 2a = 2 \sin a \cos a$$

$$= \frac{2 \tan a}{1 + \tan^2 a}$$

$$\tan 2a = \frac{2 \tan a}{1 - \tan^2 a}$$

$$\cos 2a = \cos^2 a - \sin^2 a$$

$$= 2 \cos^2 a - 1 = 1 - 2 \sin^2 a$$

$$= \frac{1 - \tan^2 a}{1 + \tan^2 a}$$

Cofunction Identities

$$\cos\left[\frac{1}{2}\pi - \theta\right] = \sin \theta$$

$$\sin\left[\frac{1}{2}\pi - \theta\right] = \cos \theta$$